

Welcome to MATH221 Linear Algebra I!

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Office Hour: TBA

Grade Distribution:

10% Attendance (10 - # absence, $0 = F$) ← University Policy

20% Quizzes (2 Quizzes)

35% Midterm

35% Final Exam

More info
in LMS

Expectations in the classroom:

- Lectures are in ENGLISH.
- Feel FREE to ASK Questions anytime during the class (you are encouraged to do so). NO

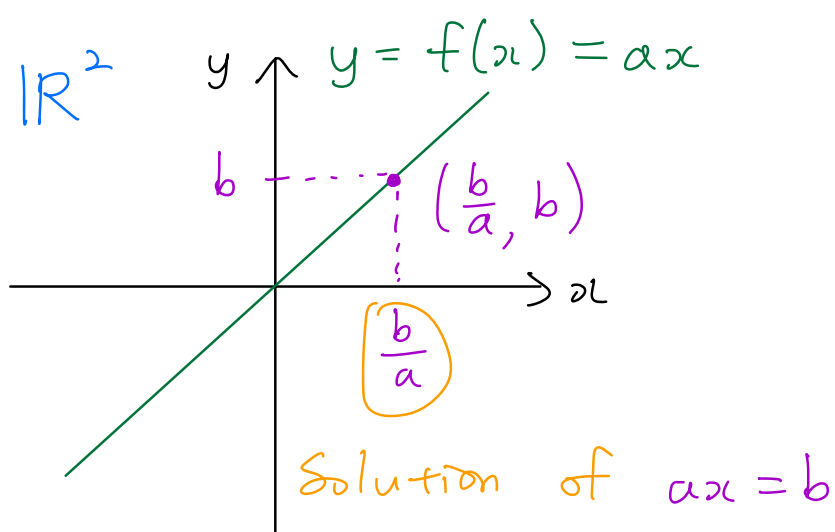
Questions are Dumb: it helps others.

- But do your best to ask in English! I will translate Questions (in Korean) into English.

So what is Linear Algebra?

Study of vector spaces and higher dimensional analog
of linear functions & equations. $(\mathbb{R}^2, \mathbb{R}^3, \text{etc})$

$\uparrow f(x) = ax$ $\uparrow ax = b$



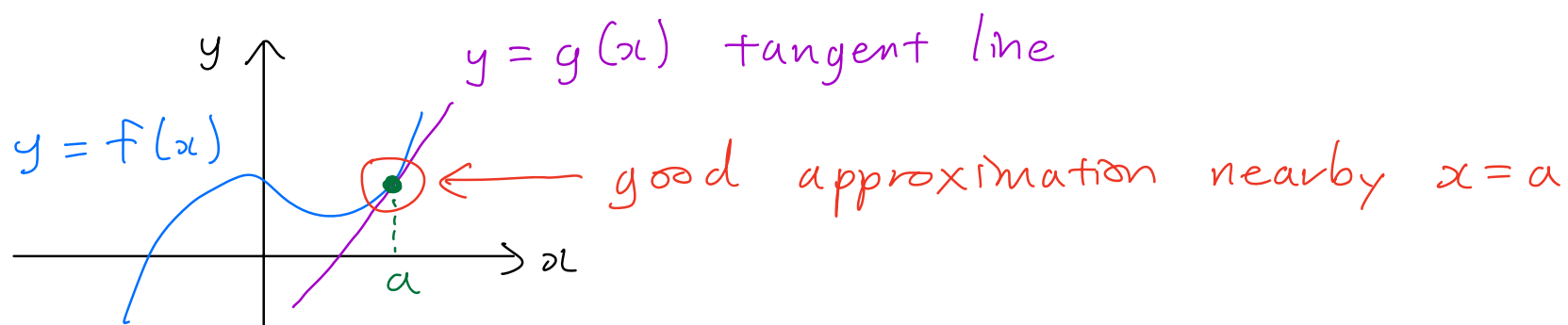
Here, a Vector refers to a point of a vector space.

For example,

- Science: an arrow $\left(\nearrow \right)$
- AI/CS: pixelated image $\left(\begin{array}{|c|c|} \hline \text{red} & \text{blue} \\ \hline \text{green} & \text{black} \\ \hline \end{array} \right)$
- Stats: a row of dataset
 $([0.7, 5.2, 4.9, 10, 8.1])$

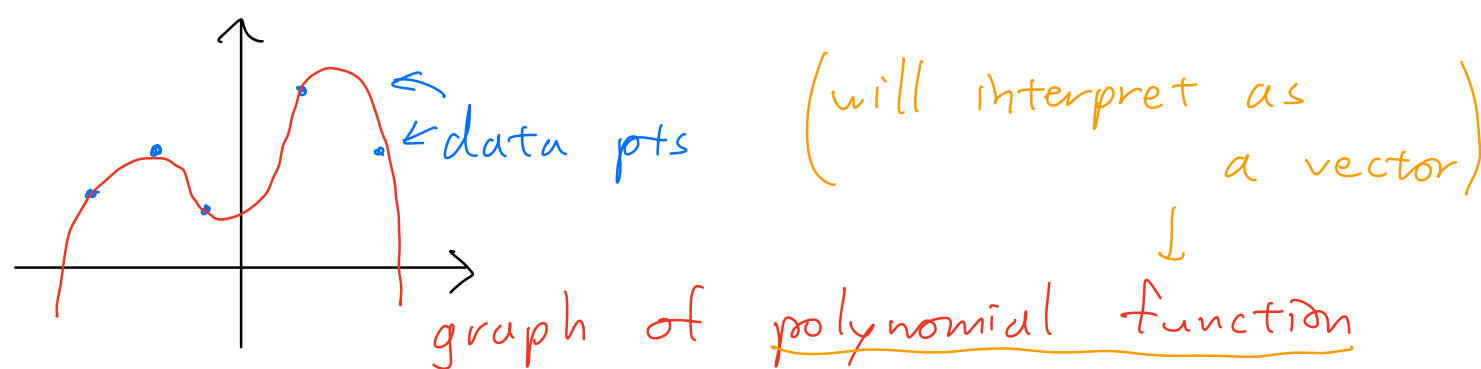
Why Linear Algebra?

- Linear Algebra, just like Calculus, is a building block of modern math.
- Linear Functions $\sim$ Complicated Functions

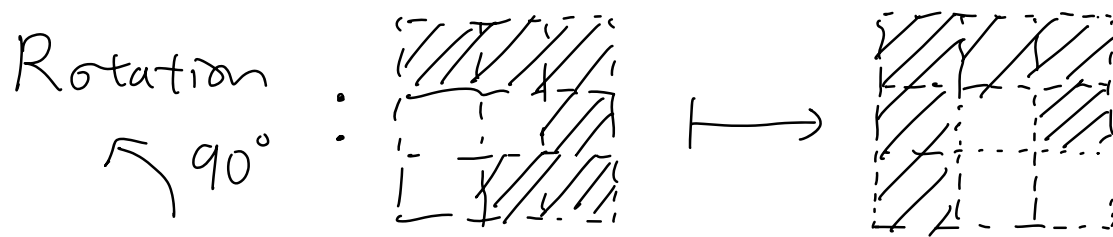


• Traditional Applications

◇ Lagrange Interpolation: find a polynomial function passing through n data points in $\mathbb{R}^2$



◇ Image Manipulation: by interpreting a pixelated image as a vector, stretch/squeeze/shear/rotate that image (via linear map)

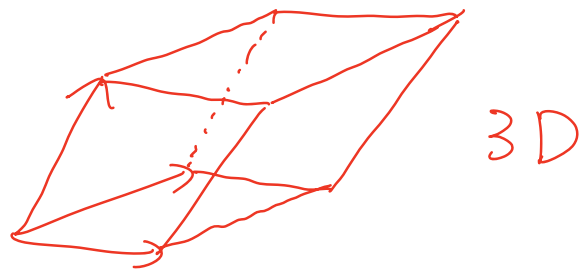
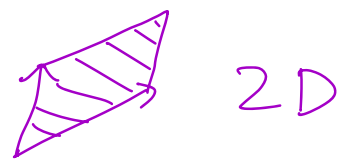


◇ Solving Systems of Linear Equations

(Algorithmically): Solve something like

$$\begin{array}{rcl} 4x + 3y & = & 9 \\ x + y & = & 2 \end{array} \quad \left| \quad \begin{array}{rcl} x + y + z & = & 4 \\ x + 2y & = & 3 \\ -y + z & = & 1 \end{array}$$

◇ (Signed) Volume of Parallelograms and Parallelepipeds



(via determinants)

This hybrid theory/computation course is about:

- Fundamentals of Linear Algebra
 - Starts with vector space instead of systems of linear equations
 - Build a proper foundation to higher math
 - Combines repetitions/patterns into a single package
- Proof-based introduction
 - Why is statement X true? Prove it!
- Course Structure:
 - (1) First 8 weeks: Heavy Theory - Building
 - (2) Last 8 weeks: Computational Applications
(with some theory)

- Unit Notes: The main textbook is used only for Practice Problems. For the content of this course, Unit Notes will be used instead. Those will be up in LMS as pdf files.